Rehab Uddin Shawon

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Research Profile

Ph.D. researcher in AI and immersive visualization with three first-author papers (two published, one accepted at IEEE ICTAI 2026) on scalable, explainable multi-agent AI for robot teams. This work targets two obstacles to deploying robot teams in warehouse automation, disaster response, and service robotics: coordination methods that break down as robot and task types grow, and AI decisions that people cannot inspect. I developed a shared-policy framework that scales to ten robot and task types, where a standard baseline fails to learn, and an explainability framework for multi-agent AI. My NSF-supported work (Award No. 2302060) included co-teaching machine learning to high school teachers. I now work in UIC's Electronic Visualization Laboratory on immersive visualization and 3D and AR/VR displays, after nearly a decade of software engineering on four globally released games with 80M+ downloads.

Key Results

- **Scalability:** at 10 robot and task types, the proposed method reached cumulative training rewards of 445.1 (heterogeneous), 366.1 (sequential), and 273.5 (collaborative) tasks, while the standard Unity ML-Agents baseline stayed near zero.
- **Generalization without retraining:** policies handled unseen task sequences with consistent accuracy (71.0–71.9%) and unseen collaboration requirements (81.0–86.2%, versus 85.6% on the trained ones).
- **Earlier framework:** Signed Type Encoding with curriculum learning significantly outperformed one-hot and no-encoding baselines ($p < 0.05$, seven independent runs), and enabled collaborative behavior that training from scratch failed to learn.
- **Explainability:** PEIM revealed team tactics such as grouping, focus fire, and fleeing in the StarCraft Multi-Agent Challenge without retraining the policies.
- **Education impact:** the ICCRE/IJMERR paper was co-authored with three high school teachers from the NSF-funded training project.

Education

University of Illinois Chicago, Chicago, IL Ph.D. Student, Electronic Visualization Laboratory (EVL) | Aug 2026 – Present | Advisor: Dr. Saeed Boorboor

Research focus: immersive visualization, 3D displays, and AR/VR.

Missouri State University, Springfield, MO M.S. in Computer Science, Data Science Certification | Jan 2025 – Jul 2026 (graduated Jul 24, 2026) | CGPA 4.00/4.00 Thesis: "Heterogeneous Robots Cooperation via Multi-Agent Reinforcement Learning" (BearWorks, Missouri State University). Committee chair: Dr. Siming Liu. Coursework: Explainable AI, Natural Language Processing, Generative AI

Jahangirnagar University, Dhaka, Bangladesh M.S. in Computer Science | Sep 2019 – Aug 2020 | CGPA 3.83/4.00

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh B.Sc. in Naval Architecture | May 2012 – Mar 2017 | CGPA 3.27/4.00

Publications

1. **Shawon, R. U.**, & Liu, S. "Priority-Aware Grid Sensing for Scalable Heterogeneous Multi-Robot Coordination." *IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026)*, Boca Raton, Florida, November 2, 2026. **Accepted.**
2. **Shawon, R. U.**, Liu, S., Cox, R., Davis, J., Walters, M., & Powers, R. "Heterogeneous Robots Cooperation via Multi-Agent Reinforcement Learning." *International Journal of Mechanical Engineering and Robotics Research (IJMERR)*, vol. 15, no. 3, 2026. Presented at ICCRE 2026, Japan. Supported by the National Science Foundation (Award No. 2302060) through the ROSE project; co-authored with participating high school teachers.
3. **Shawon, R. U.**, Liu, S., & Siddiqua, A. "Explainable Reinforcement Learning for Multi Agent Systems." *2025 IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, IEEE, 2025, pp. 1339–1343. DOI: 10.1109/ICTAI66417.2025.00194.

Research Experience

Graduate Research Assistant, Missouri State University | Jan 2025 – Jul 2026 Advisor: Dr. Siming Liu

- Designed a shared-policy MARL framework for heterogeneous robot teams using a priority-aware grid sensor that embeds task relevance directly into each robot's observation. It scales to up to 10 robot and task types.
- Evaluated across heterogeneous, sequential, and collaborative task settings in Unity (126 training runs over 3-, 5-, and 10-type configurations). At 10 types, the proposed method reached cumulative training rewards of 445.1 (heterogeneous), 366.1 (sequential), and 273.5 (collaborative), while the default Unity ML-Agents grid sensor stayed near zero.
- Developed generalized formulations for sequential tasks of arbitrary length and order and for collaborative tasks with dynamic robot requirements. Policies generalized to unseen task sequences and requirement vectors without retraining.
- Built a training pipeline for heterogeneous robots performing unique, sequential, and collaborative tasks, including policy transfer from single-task learning to multi-task collaboration.
- Developed PEIM (Policy Explanation in MARL), a model-agnostic framework that explains individual and collaborative agent decisions in MARL; tested on the StarCraft Multi-Agent Challenge (SMACv2) and published at IEEE ICTAI 2025.
- Created a Meta-Ensemble Explainable AI framework for image data that combines multiple XAI methods (LIME, SHAP, Grad-CAM, LRP, SmoothGrad) and learns to merge their outputs into more reliable instance-level explanations.

Research Project Assistant, Jahangirnagar University | Sep 2019 – Aug 2020 Advisor: Dr. Mushfique Anwar

- Developed image editing software in Unity and C# applying digital image processing techniques.

Outreach and Teaching

NSF-funded ROSE Project (Award No. 2302060), Missouri State University

- Co-taught and trained high school teachers in machine learning with Dr. Siming Liu, so they could bring ML into their own classrooms and schools.
- Conducted research with the program and co-authored the resulting journal paper (IJMERR 2026).

Graduate Teaching Assistant, Missouri State University | Jan 2025 – Jul 2026

- Led lab sessions for 60+ students in data structures and algorithms.

- Provided 6 hours per week of tutoring and graded 100+ assignments per semester.

Industry Experience

Senior Game Programmer, Kolpoverse Studios, Dhaka | Nov 2022 – Nov 2024

- Engineered high-performance Unity (C#) systems, optimizing rendering, physics, and memory, and implemented IK, cloth, custom physics joints, and mesh manipulation.

Game Programmer, Alpha Potato Games, Dhaka | Sep 2019 – Oct 2022

- Built core gameplay systems for 30+ Unity prototypes and 4 globally released titles with 80M+ downloads. Integrated analytics SDKs and delivered Android/iOS build pipelines.

Game Programmer, Mindfisher Games, Dhaka | Jun 2017 – Aug 2019

- Built 20+ Unity prototypes and shipped multiple internal and soft-launch projects using Git and Agile delivery.

3D Animator, Portbliss Games, Dhaka | Apr 2015 – May 2017

- Built animation pipelines and Unity animation controllers for *Heroes of 71* (1M+ downloads).

Technical Skills

- **Languages:** Python, C#, C++, Java
- **ML and data:** PyTorch, TensorFlow, Unity ML-Agents, Scikit-learn, Pandas, NumPy, Apache Spark; explainability tools (LIME, SHAP, Grad-CAM, LRP, SmoothGrad)
- **Environments and tools:** Unity, Unreal Engine, SMACv2, PyGame, Git/GitHub, MySQL, Firebase, Power BI, Tableau
- **English proficiency and testing:** IELTS 7.5, GRE 308